

Determining M_n/M_p for ^{17}C and Testing Ab Initio Theory

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I. MOTIVATION

In the past few decades, significant experimental and theoretical work has been devoted to the neutron-rich carbon isotopes. Many of these studies have focused on the evolution of the shell-model orbitals as the isotopes become more neutron rich. For example, evidence has been presented for a $Z = 6$ shell gap that persists along the isotopic chain [1], although a more recent study has brought this result into question [2]. Conversely, the near-degeneracy of the $\nu 1d_{5/2}$ and $\nu 2s_{1/2}$ orbitals in the neutron-rich carbon isotopes results in the disappearance of the $N = 14$ sub-shell gap [3]. This is opposite to the effect that is observed in the neutron-rich oxygen isotopes, which exhibits both robust $N = 14$ and $N = 16$ sub-shell gaps [4]. This difference between the carbon and oxygen isotopes highlights the importance of the interaction between the core and valence nucleons.

A. Trends in $B(E2)$

Multiple studies have determined the $B(E2)$ value of the first 2^+ state in ^{16}C via lifetime measurements [5–8]. While discrepancies exist, the latest evaluation of $B(E2)$ values lists the recommended ^{16}C $B(E2)$ value at $0.00179(20) \text{ e}^2\text{b}^2$ or $1.50(17) \text{ W.u}$ [9]. In terms of W.u., this value is a factor of 2 smaller than its isotone, ^{18}O , suggesting a quenching of the quadrupole transition strength. This quenching persists in ^{18}C . Two lifetime measurements have been performed and agree within uncertainty [7, 10], leading to a recommended $B(E2)$ value of $0.00168(^{+23}_{-15}) \text{ e}^2\text{b}^2$ or $1.20(^{+16}_{-11}) \text{ W.u}$ [9]. Again, in terms of W.u. this is smaller than the $B(E2)$ of its isotone, ^{20}O .

However, the $B(E2)$ value appears to increase in ^{20}C . One lifetime measurement has been performed, resulting in a $B(E2)$ value of $0.0038(^{+17}_{-8}) \text{ e}^2\text{b}^2$ or $2.3(^{+10}_{-5}) \text{ W.u}$. [11]. The authors explain this increase as a consequence of the decreased energy gap between the $\pi 1p_{1/2}$ and $\pi 1p_{3/2}$ orbitals that occurs due to the interaction with the $\nu 1d_{5/2}$ orbital. Macchiavelli *et al.* furthered this explanation by employing a seniority-inspired scheme to describe the neutrons of ^{16}C , ^{18}C , and ^{20}C [12]. Their phenomenological analysis studied wave functions, in particular, what fraction of the amplitude correspond to a proton excitation into the $1p_{1/2}$ orbital. The proton amplitude was relatively small ($\sim 12\%$) in both ^{16}C and ^{18}C , but much larger ($\sim 30\%$) in ^{20}C .

While this phenomenological analysis is at odds with Ref. [1], which suggests a robust $Z = 6$ closed shell across the carbon isotopic chain, it is supported by the results of proton knockout reactions on ^{17}N , ^{19}N , and ^{21}N [2]. The knockout reaction work shows a relatively constant proton amplitude in the 2^+ states of ^{16}C and ^{18}C and then an increase in the amplitude in ^{20}C . However, the increase in the proton amplitude is not as large as implied by the seniority scheme, and the experimentally-determined proton amplitudes for ^{16}C and ^{20}C nearly agree within uncertainty. While suggestive that the proton amplitude increases, it is by no means definitive.

B. Trends in M_n/M_p

To aid in understanding of the neutron-rich carbon isotopes, experimental results complementary to $B(E2)$ values are needed. While $B(E2)$ values are only sensitive to the proton contributions to nuclear transitions,

M_n/M_p values provide additional information on the neutron contribution to nuclear transitions. M_n/M_p values can probe the degree of coupling between valence nucleons and the core. Strong coupling between the valence nucleons and the core is expected in collective nuclear states, and in a simple homogeneous, liquid-drop model the M_n/M_p value should be close to N/Z . On the other hand, weak coupling is expected in states that are more “shell-model like.” For these states the quadrupole transition is driven by the valence nucleons. For nuclei with valence neutrons, this results in M_n/M_p values that are larger than N/Z . As such, the value of $(M_n/M_p)/(N/Z)$ is often reported. Multiple studies have been performed to determine the M_n/M_p values of the neutron rich carbon isotopes and to determine the degree of core-valence decoupling in these exotic nuclei.

M_n/M_p values for the neutron-rich carbon isotopes have been determined through combinations of lifetime measurements and inelastic scattering measurements. One measurement, performing inelastic scattering of ^{15}C on a deuteron target reported a $(M_n/M_p)/(N/Z)$ value of ~ 2.4 , suggesting valence neutron decoupling from the ^{14}C core [13]. Multiple studies have been performed to determine the M_n/M_p value of ^{16}C [6, 7, 14–16]. The most recent resulted from a measurement of ^{16}C inelastic scattering on both protons and deuterons and reported $(M_n/M_p)/(N/Z) = 1.2$. While not suggestive of valence neutron decoupling, the authors also give an overview of previous measurements, where $(M_n/M_p)/(N/Z)$ values range from $\sim 1.4 - 2$.

Currently, no studies have reported M_n/M_p values for ^{17}C , ^{18}C , or ^{19}C . For ^{20}C , combining inelastic scattering measurements from Ref. [14] with the $B(E2)$ value reported by Ref. [11] yields $(M_n/M_p)/(N/Z) = 1.2$. This lower M_n/M_p value suggests a re-coupling of the valence neutrons to the ^{14}C core, which is consistent with the increased proton amplitude in the 2^+ state of ^{20}C .

While discrepancies exist in the experimental data, there appears to be an increase in the $B(E2)$ value between ^{16}C and ^{20}C which then results in a decrease in the M_n/M_p value. Recent no-core shell model (NCSM) calculations have also been performed, and results may help shed light on the structural evolution of the neutron-rich carbon isotopes. Fig. 1 shows the results of NCSM calculations of $(M_n/M_p)/(N/Z)$. In the even isotopes, there is a distinct trend across the isotopic chain, going from a large M_n/M_p value for ^{16}C to a smaller M_n/M_p value for ^{20}C . This trend suggests significant valence neutron decoupling in the lighter isotopes and a return to collectivity as neutrons are added. This is consistent with the measured M_n/M_p values for ^{16}C and ^{20}C previously discussed.

Fig. 2 shows the ground-state orbital occupations extracted from NCSM calculations [17]. The NCSM calculations show only a moderate increase in the proton $1p_{1/2}$ contribution going from ^{16}C to ^{20}C , at odds with the sudden increase argued for in Ref. [12] and possibly seen in Ref. [2]. In fact, the seniority scheme put forward in Ref. [12] would seem to suggest a value for M_n/M_p that is much less than 1, at odds with a collective model and in contrast to both NCSM calculations and experimental results.

An alternative explanation for the increase in the $B(E2)$ values (and subsequent decrease in M_n/M_p) is given by AMD calculations [18]. These calculations show the protons to be slightly oblate across the isotopic chain. On the other hand the neutrons are prolate in ^{16}C and transition to oblate in ^{20}C . If the quadrupole transitions are primarily driven by valence neutrons, the misalignment in the proton and neutron symmetry axes is then what leads to the large $(M_n/M_p)/(N/Z)$ in ^{16}C and the re-alignment of the symmetry axes (and restoration of collectivity) is what leads to the decreased $(M_n/M_p)/(N/Z)$ value in ^{20}C . This picture is supported by SU(3) decompositions of NCSM wavefunctions. The proton shape is consistently oblate, and the neutron shape gradually transitions from prolate at ^{16}C to oblate at ^{20}C .

In the odd isotopes, the picture is more complicated. A large $(M_n/M_p)/(N/Z)$ value is seen in ^{15}C . This is consistent with experimental information and the predicted $(M_n/M_p)/(N/Z)$ value of ^{16}C . However, both ^{17}C and ^{19}C have a triplet of $J^\pi = 1/2^+, 3/2^+, 5/2^+$ states at low excitation energies. For these states, the

$(M_n/M_p)/(N/Z)$ value for both the $1/2^+ \rightarrow 3/2^+$ and $5/2^+ \rightarrow 3/2^+$ transitions are shown in Fig. 1. In ^{17}C , the $(M_n/M_p)/(N/Z)$ for the $5/2^+ \rightarrow 3/2^+$ transition is similar to that of ^{16}C , while for the $1/2^+ \rightarrow 3/2^+$ transition the $(M_n/M_p)/(N/Z)$ value is close to 1. For ^{19}C , it is the opposite, with the $(M_n/M_p)/(N/Z)$ value of the $1/2^+ \rightarrow 3/2^+$ transition larger than that of the $5/2^+ \rightarrow 3/2^+$ transition.

The case of ^{17}C is particularly interesting. The $3/2^+$ ground state of ^{17}C has been shown to consist of a $^{16}\text{C}(2^+) \otimes \nu 1d_{5/2}$ component with a smaller $^{16}\text{C}(2^+) \otimes \nu 2s_{1/2}$ component, as well a small $^{16}\text{C}(0^+) \otimes \nu 1d_{3/2}$ component [19]. Only two bound, excited states are present in ^{17}C , at energies of 217 keV and 332 keV, with spin-parity assignments of $1/2^+$ and $5/2^+$, respectively. A recent transfer study has revealed these states to be primarily $^{16}\text{C}(0^+) \otimes \nu 2s_{1/2}$ and $^{16}\text{C}(0^+) \otimes \nu 1d_{5/2}$, respectively [20].

The structure of ^{17}C outlined above would seem to imply a similar M_n/M_p value for transitions $1/2^+ \rightarrow 3/2^+$ and $5/2^+ \rightarrow 3/2^+$. Furthermore, this implied $(M_n/M_p)/(N/Z)$ value should be similar in magnitude to ^{16}C . This is consistent with NCSM calculations for the $5/2^+ \rightarrow 3/2^+$ transition. However, the low $(M_n/M_p)/(N/Z)$ value for the $1/2^+ \rightarrow 3/2^+$ transition suggests an increase in collectivity. This is surprising, as Ref. [20] proposes the $1/2^+$ state as a potential halo state, and as such additional valence neutron decoupling due to the halo nature of the $1/2^+$ state could be expected to lead to a larger M_n/M_p for the $1/2^+ \rightarrow 3/2^+$ transition compared to the $5/2^+ \rightarrow 3/2^+$ transition. A possible explanation for the dramatic difference between the M_n/M_p values for the two transitions in ^{17}C could be differing neutron shapes.

As noted previously, no experimental studies have been performed to determine M_n/M_p in ^{17}C . Therefore, to further our understanding of the neutron-rich carbon isotopes and test ab-initio nuclear theory, we propose to perform the first experimental determination of M_n/M_p in ^{17}C by measuring inelastic scattering of ^{17}C on both proton and deuteron targets.

II. EXPERIMENTAL DETAILS

A. Determining M_n/M_p

In order to determine the M_n/M_p value of the first two excited states in ^{17}C , we will measure differential cross sections of $^{17}\text{C}(d,d')^{17}\text{C}$ and $^{17}\text{C}(p,p')^{17}\text{C}$ using the AT-TPC and SOLARIS. The ratio of the quadrupole matrix elements can be described as $M_n/M_p = (N\delta_n)/(Z\delta_p)$, where $\delta_{n(p)}$ is the neutron(proton) deformation length [21].

In general, the matter deformation parameter can be obtained from inelastic scattering using a hadron probe h by comparing the experimental differential cross section to DWBA cross sections using a rotational model. Specifically, $(d\sigma/d\Omega)_{exp} = \beta_{hh'}^2 (d\sigma/d\Omega)_{DWBA}$ where $\beta_{hh'} = \delta_{hh'}/(r_0 A^{1/3})$ is the deformation determined from the probe. Then, $\frac{\delta_{hh'}}{\delta_p} = \frac{1+(b_n^h/b_p^h)(M_n/M_p)}{1+(b_n^h/b_p^h)(N/Z)}$, where b_n^h/b_p^h is the ratio of the interaction strength between the hadron probe h and the neutron and proton [21].

Because $E2$ transitions are purely electromagnetic, the proton deformation length, δ_p can be extracted from $B(E2)$ values. Lifetime measurements exist for the first two excited states in ^{17}C [22]. However, these transitions are purely $M1$ in nature, and therefore the lifetimes can not be used to determine the $B(E2)$ value. Therefore, it is necessary to measure the matter deformation via inelastic scattering with two different hadron probes, and use the ratio of these two probes to eliminate the dependence on δ_p and extract the M_n/M_p value. For this work, the two probes used will be protons and deuterons.

B. Experimental Method

We will measure the inelastic scattering cross section with SOLARIS and the AT-TPC, utilizing both hydrogen and deuterium gas in the AT-TPC. A diagram of SOLARIS is shown in Fig. 3. A re-accelerated ^{17}C beam will be delivered to SOLARIS and the AT-TPC on the ReA6 beamline at 8.37 MeV/u and 940 pps [23]. The only expected contaminant is ^{17}O , but the amount of contaminant expected was not able to be estimated. However, as the Z of the two beam components differs by 2, the AT-TPC is expected to distinguish between ^{17}C and ^{17}O .

The AT-TPC will act as the target and also detect the scattered protons and deuterons. By coupling the AT-TPC and SOLARIS, the reactions will take place in a strong, uniform magnetic field and the ejectiles will undergo cyclotron motion. These helical particle trajectories will be tracked by the AT-TPC, leading to clean separation between the scattered ejectiles of interest and ejectiles from other reaction channels such as (d,p), (p,d), (d, ^3He), etc. We also note that, while not the primary motivation for the proposed experiment, these other reaction channels will provide a wealth of additional information regarding the structure of the neutron-rich carbon isotopes.

The coupling of SOLARIS with the AT-TPC has previously been used to measure the inelastic scattering of ^{14}C on protons [24]. In this study, a ^{14}C beam of 12.4 MeV/u was used and the AT-TPC was filled with 300 torr of pure H_2 gas. With this configuration, only 25 hours of beam time was required to measure a differential cross section.

DWBA calculations were performed using FRESKO [25] to compare the differential cross sections for inelastic scattering to the first 2^+ state in ^{14}C and the first $5/2^+$ state in ^{17}C . The differential cross sections were found to be comparable. Fig. 4 of Ref. [24] shows the excitation energy spectra in the top left and the inelastic differential cross section to the first 1^- state in ^{14}C . Clear structure can be seen in the differential cross section. Furthermore, as can be seen in the excitation energy plot, the inelastic scattering to the first 2^+ state in ^{14}C is nearly a factor of 4 stronger than the inelastic scattering to the first 1^- state. Therefore, while the ^{17}C beam rate is expected to be a factor of 2 lower than ^{14}C , one day of beam time with each target gas is expected to provide enough statistics to measure the differential cross section curves to both the $1/2^+$ and $5/2^+$ states. We also note that while the separation between the two states is only 130 keV, the excitation energies between the two states is well known. Therefore, even with the 300 keV energy resolution achievable in the AT-TPC, we expect to be able to fit a doublet in order to extract counts from the individual states.

C. Beam Time Requested

Due to the thickness of the gas target offered by the AT-TPC, along with the excellent detection efficiency, we request 2 days of beam time. One day will be devoted to proton scattering, where the AT-TPC will be filled with hydrogen gas. The other day will be devoted to deuteron scattering, where the AT-TPC will be filled with deuterium gas. Changing gases in the AT-TPC is a standard practice and is not expected to take a considerable amount of time. This request does not include time for beam tuning. Based on the previously-mentioned inelastic scattering work of Ref. [24], we expect this to result in ~ 1000 counts of scattered protons and a similar amount for deuterons. Similar statistics were achieved in Ref. [13], and in that work the matter deformation was determined with $\sim 10\%$ uncertainty. We therefore assume similar uncertainty in this measurement, which would result in an M_n/M_p value with $\sim 15\%$ uncertainty. This is enough to distinguish whether or not the M_n/M_p suggests strong or weak valence neutron coupling.

III. FIGURES AND DIAGRAMS

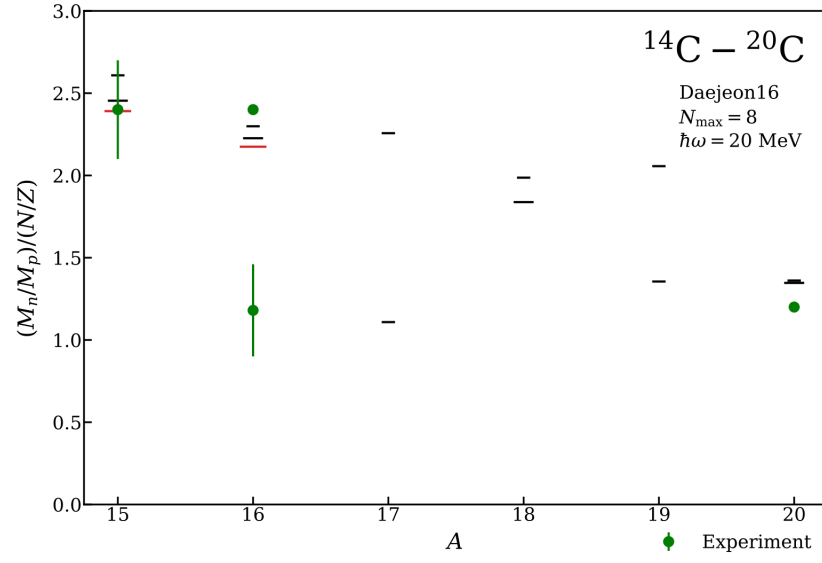


FIG. 1. M_n/M_p calculations from NCSM calculations.

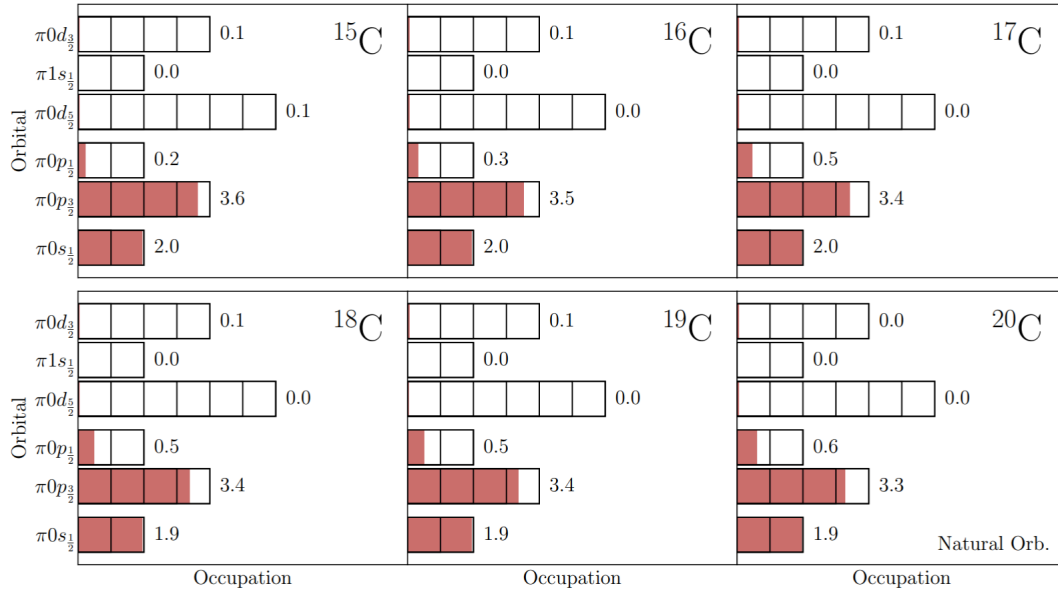


FIG. 2. Proton occupation numbers from NCSM calculations [17]

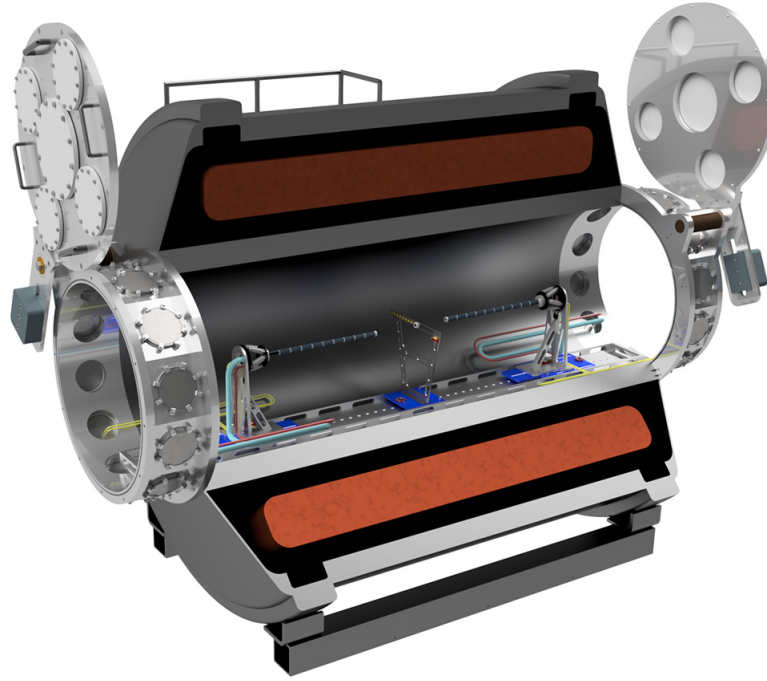


FIG. 3. Diagram of SOLARIS [26]

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